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Does a Vision Transformer really beat CNNs on pediatric pneumonia X-rays?

A reality check of Singh et al., "Efficient pneumonia detection using Vision Transformers on chest X-rays", Scientific Reports 14:2487 (2024)

The claim, in one sentence

A pretrained Vision Transformer (DeiT-Base) detects pneumonia in children's chest X-rays more accurately than any CNN: 97.6 % accuracy, against 76–94 % for fifteen CNNs.

Why it matters for us

Same dataset and task as our Assignment 1, where a fair protocol gave ResNet-50 the best AUROC (0.971) but only 86 % accuracy. Either the transformer is better, or the 97.6 % is the protocol. Our pipeline can tell the two apart.

What we found

The 97.6 % reproduces under their split (we get 98.0 %), and a plain ResNet-50 gets 97.2 % under the same split. Under a patient-grouped split both fall to 82–86 %. The number is the split, not the transformer.

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The paper: problem, claim, method, evidence

Problem

Classify a child's frontal chest X-ray as NORMAL or PNEUMONIA. Kermany et al. 2018 dataset: 5,856 images from Guangzhou Women and Children's Medical Center, 73 % pneumonia.

Claim

An ImageNet-pretrained Vision Transformer (DeiT-Base/16, 224 px) is more accurate than CNN architectures: 97.61 % accuracy, sensitivity 0.949, specificity 0.981, AUC 0.966.

Method

Random 80/10/10 split (4,684 / 586 / 586). DeiT-B fully fine-tuned: Adam, LR 1e-5 × 0.995 per epoch, batch 16, 30 epochs, cross-entropy. No augmentation described.

Evidence

One run, one split. Accuracy with a Wald 95 % CI, a confusion matrix, ROC and PR curves, and Table 6: fifteen CNNs at 76–94 % accuracy.

How they support the claim (their Table 6, selected rows)

Architecture	Accuracy %	Trainable params	Non-trainable params
ResNet-50	82.37	200,706	23,587,712
DenseNet-121	91.82	100,354	7,037,504
ResNet-152V2 (best CNN)	92.94	200,706	58,331,648
Vision Transformer (proposed)	**97.61**	**85,800,194**	**0**

Supporting papers

Dack & Dai, ICCV Workshops 2025: dataset bias in chest X-rays; pixel- and patch-shuffling tests on lung-only images.

Khadidos et al., arXiv 2026 (preprint): EfficientNet-B0 / DenseNet-121 on the official test folder reach 85 % / 80 % accuracy, AUC 0.97–0.98, i.e. our Assignment 1 range.

How they tested it, and where the evidence is weakest

200,706

trainable parameters in their ResNet-50 (of 23.8 M). The CNN backbones were frozen; the ViT had all 85.8 M trainable

0 words

about patients. "80 % training, 10 % test, 10 % validation" is the whole description of the split

330 ≠ 160

pneumonia test images: Table 3 says 330 / 256, the confusion matrix implies 160 / 426

1 run

no seed, no cross-validation, no variance; test accuracy tracked every epoch

1 · Baselines not fine-tuned (Table 6)

"All hyperparameters were kept the same", but the models were not trained the same way: frozen CNN backbone + small head vs a fully fine-tuned transformer. The table measures fine-tuning vs feature extraction, not ViT vs CNN. A 'Ref.' column beside every baseline leaves open whether some numbers were run at all.

2 · Split ignores patients and samples the training pool as test. 1,155 children have several scans (up to 30). In A1 a naive by-image split put 61 % of held-out images beside a scan of the same child. And the official test folder is a different collection: every A1 model scored ≥ 0.99 AUROC on validation but 0.90–0.97 on it. A random 10 % never faces that shift.

3 · Numbers that do not reconcile, and thin reproducibility

F-score 0.952 (Table 5) vs 0.950 (Table 6); cross-entropy in one section, binary cross-entropy in another; stated CI 96.2–98.9 % vs 96.4–98.9 % from their own Eq. 7; 5,863 images in the text, 5,856 in the tables.

Limitations: one sentence ("data scarcity and the need for real-world validation"). No code, no split files, no seeds.

Verdict so far: the numbers support "a fine-tuned DeiT-B reaches ~98 % on a random image split". They do not support "ViT outperforms CNNs". That is a testable gap.

Reality check: their model under our protocol, their protocol on our models

One pipeline, nothing else changes

Assignment 1 code, unchanged: grayscale → pad-to-square → 224 px, same augmentation, AdamW, 12 epochs, checkpoint by best validation AUROC, seeds 42 and 43, one Colab T4.

Two additions: the paper's model, and a switch for the paper's split.

Their model, faithfully

timm deit_base_patch16_224, ImageNet weights, fully fine-tuned, LR 1e-5 as in the paper. 85.8 M parameters, 9 min per run.

Closest approximation possible without their code; the remaining gaps (30 vs 12 epochs, plain resize, Adam vs AdamW) are listed in the report.

Their protocol, faithfully

Pool the 5,824 de-duplicated images, random label-stratified 80/10/10 split by image, no patient grouping: 4,659 / 582 / 583.

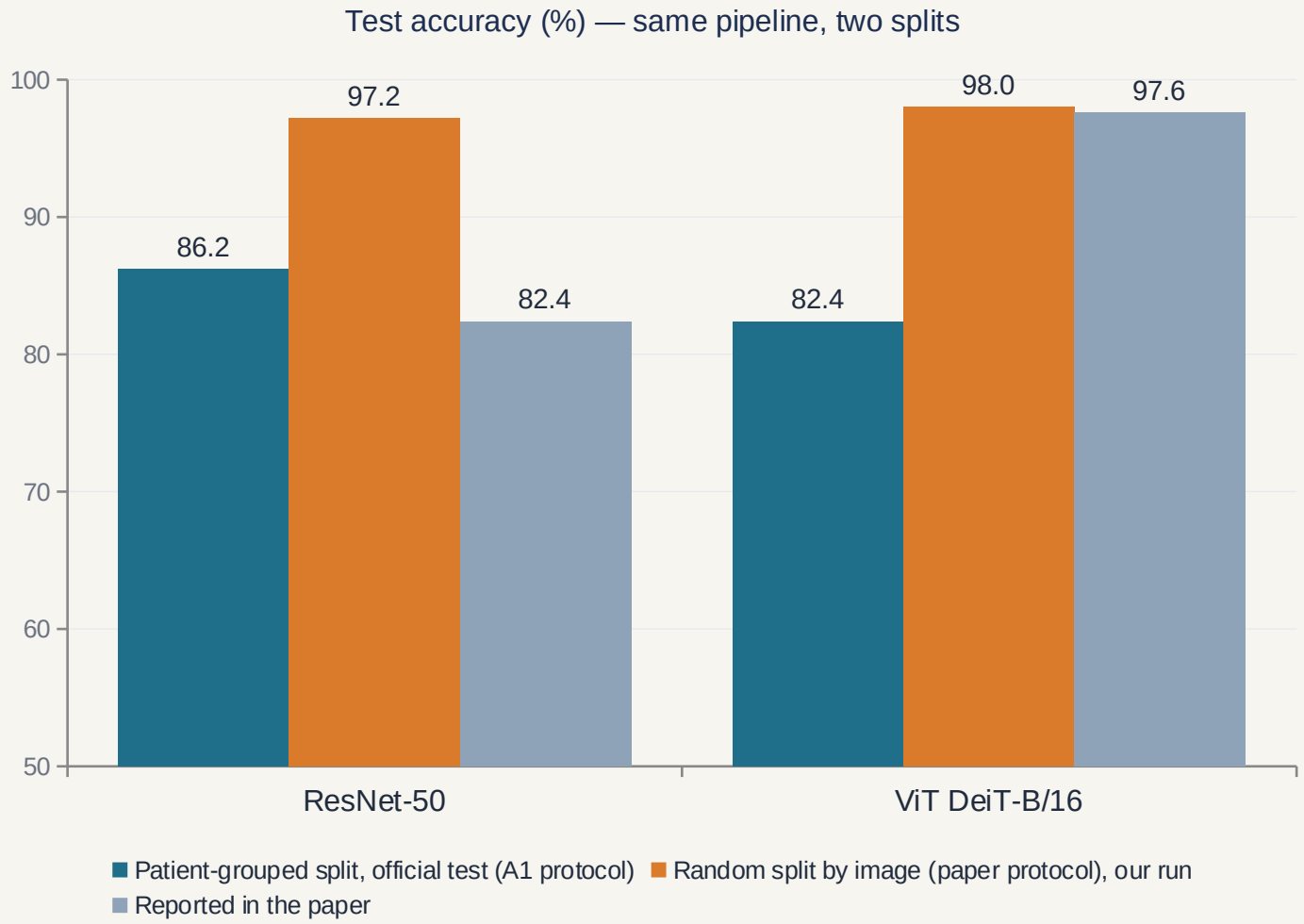
Measured leakage: 336 of 583 test images (58 %) share a patient with the training set. Under our split: 0.

What we ran (mean of seeds 42 and 43)

	A1 protocol: patient-grouped split, official test folder (n = 618)	Paper protocol: random split by image (n = 583)
ViT DeiT-B/16 (paper's model)	run · new	run · new
ResNet-50 (A1 winner)	A1 result	run · new
DenseNet-121, small CNN, logistic regression, majority	A1 results	not run (time)

Honest note: our first image-split run was invalid (the split had pooled the dropped duplicate rows of the index). We caught it by inspecting the outputs, fixed the code, added a guard, and re-ran all four runs; the invalid files were discarded and the fix is in the commit history.

Result: the 97.6 % reproduces, and a plain ResNet-50 gets the same number



98.0 %

our ViT under their split (theirs: 97.6 %). Specificity 0.97 vs their 0.98. Reproduced.

97.2 %

our fine-tuned ResNet-50 under their split (they report 82.4 % with a frozen backbone)

82.4 %

the same ViT under the patient-grouped split and the official test folder (specificity 0.53)

0.973

ViT AUROC under the fair protocol vs 0.971 for ResNet-50; seed spread 0.001. A tie.

Cost of the tie

85.8 M vs 23.5 M parameters (3.7×), 9.2 vs 4.4 min training (2×), lower accuracy and specificity.

Why: two properties of the data explain the whole gap

Property 1 · Patient leakage (58 % of their test set)

Under the paper's split, 336 of 583 test images have a scan of the same child in training. Both models memorise children and are rewarded for it: specificity jumps from 0.53–0.64 to 0.97–0.98. Under patient grouping, leakage is 0 and the numbers fall back to the A1 range.

Property 2 · The official test folder is a different collection

Every model, ViT included, reaches 0.999 AUROC on our patient-grouped validation set, then 0.90–0.97 on the official test folder. The drop is a collection shift, not under-training. A random 10 % of the pool never faces that shift, so their test is easier by construction.

Plus one property of their setup

Frozen CNN backbones. Fully fine-tuned, ResNet-50 and ViT are within one point under their protocol and within 0.002 AUROC under ours. Architecture is not what moved the number; the evaluation protocol is.

Verdict

The claim does not hold up. What holds up: "a fine-tuned DeiT-B reaches ~98 % on a random image split", and, more usefully, "on this dataset the evaluation protocol dominates the architecture". The paper's own figures, re-run fairly, say so.

What we would do next, and our question for another group

Close the remaining gaps

One run each: their 30-epoch budget and plain resize under the patient split (to confirm the ranking does not move), and a third seed. Run the image split on the three remaining A1 models to show the inflation is model-independent.

Test the transformer where it could matter

An external pediatric test set (e.g. VinDr-PCXR) instead of this dataset's own folder, and a train-size sweep: ViTs are known to need more data than CNNs, and 4,400 images is small.

For the course project

The same three checks (patient grouping, collection shift, baseline parity) become a checklist we apply to any benchmark claim before building on it.

Our question for another group

"In the paper you critiqued, were the baselines trained with the same budget and the same trainable parameters as the proposed model, and how would you check that from the paper alone?"